

European Girls Math Olympiad 2023 — P1/3

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There are $n \geq 3$ positive real numbers $a_1, a_2, \dots, a_n$. For each $1 \leq i \leq n$ we let $b_i = \frac{a_{i-1} + a_{i+1}}{a_i}$ (here we define a_0 to be a_n and a_{n+1} to be a_1). Assume that for all i and j in the range 1 to n , we have $a_i \leq a_j$ if and only if $b_i \leq b_j$.

Prove that $a_1 = a_2 = \dots = a_n$.

Solution

Suppose, for the sake of contradiction, that there exists $1 \leq i < j \leq n$ such that $a_i < a_j$. Let a_M be the largest number in (a_n) . Let a_m be the smallest number in (a_n) . Note that b_m is the smallest number in (b_n) and that b_M is the largest number in (b_n) , because of the fact that $a_i \leq a_j \iff b_i \leq b_j$. Now observe that

$$b_M = \frac{a_{M-1} + a_{M+1}}{a_M} \leq 2, \quad b_m = \frac{a_{m-1} + a_{m+1}}{a_m} \geq 2$$

and so $b_m = b_M = 2$, which means that $b_1 = b_2 = \dots = b_n = 2$. Now since we have equality over all b_i the statement $b_i \leq b_j \iff a_i \leq a_j$ is true for all pairs of integers (i, j) . That means that we have $a_i \leq a_j$ and $a_i \geq a_j$ and so $a_i = a_j$ again for all pairs of integers (i, j) . Finally this means that $a_1 = a_2 = \dots = a_n$ which is a contradiction. So indeed $a_1 = a_2 = \dots = a_n$. ■